

Bayesian Technosignature Detection: Adapting Gravitational-Wave Inference Methods for Radio SETI

Abstract

Radio technosignature searches produce millions of candidate signals per survey, yet the classification pipelines determining which candidates warrant a follow-up has remained fundamentally unchanged for decades, with signal-to-noise thresholding, heuristic radio frequency interference (RFI) filters, and manual visual inspection often used. These approaches do not quantify the confidence of individual classifications, neither do they produce probabilistic principled scoring for follow-up prioritisation. I propose a Bayesian multi-hypothesis inference framework, that addresses these limitations by drawing directly on methods developed for gravitational-wave detection, a field that faced structurally identical problems before the first LIGO detections. The designed framework is positioned as a downstream complement of existing machine learning pipelines, not a replacement, that is, machine learning finds the candidates and Bayesian inference determines what they mean.

1 Introduction

The State of the Art and Its Limits

The Breakthrough Listen (BL; Worden et al., 2017) Initiative represents the most systematic technosignature search in history with thousands of hours of telescope time dedicated to the search. Its null results set the most stringent upper limits to date on the prevalence of artificial transmitters (e.g. Price et al., 2020; Choza et al., 2024). Yet the data analysis pipeline, (turboSETI; Enriquez et al., 2017) that underpins these results rests on a foundation that has not substantially changed since the earliest SETI surveys: deterministic thresholding, heuristic RFI filtering and manual candidate inspection. Although machine learning (ML) techniques including semi-supervised methods (e.g. Ma et al., 2023) and anomaly detection techniques (e.g. Pardo et al., 2025) which drastically reduce the false positive even rates have now been employed, the operational consequence is severe. A recent analysis of 10^{11} spectrograms required human vetting of 20,000 candidates (Pardo et al., 2025). Yet, even after this classification using ML algorithms or by visual inspection, the final result remains probabilistically unquantified. When a genuinely anomalous candidate such as the Breakthrough Listen Candidate 1 (BLC-1; Smith et al., 2021) survives all automated filters and visual inspection, the community has no principled way to ask: *how confident are we this is extraordinary?* Answering this question often requires months of expert analysis (Sheikh et al., 2021).

The Missing Layer

The gravitational-wave (GW) community confronted an analogous problem in the years before the first LIGO detection. The pipelines relied on power-searches and heuristic vetos (e.g. Abbott et al., 2005; Abadie et al., 2010), with candidates often requiring extensive manual follow-ups. The solution was not to replace the pipelines but to place a rigorous probabilistic layer based on Bayesian parameter estimation on top of them (Veitch & Vecchio, 2008; Veitch et al., 2015), where Bayesian evidence and model comparison are computed. The result was a framework in which Bayesian inference over a probability distribution $\mathcal{P}(\theta|\mathcal{D})$, where θ are the model parameters and $\mathcal{D}$ the data became the standard in the GW community (Abbott et al., 2019a). The use of this probabilistic approach in the GW community enabled quantified uncertainty, principled follow-up prioritisation (Martins et al., 2026) and population inference from an ensemble of detections (Abbott et al., 2019b). **We argue that radio SETI is at exactly the same inflection point, and propose to transplant this methodology.** The specific adaptations are non trivial since SETI has no matched waveform family or known source population, however, the conceptual machinery transfers directly as demonstrated in section 2.

Why Now?

The urgency of this work is driven by the imminent scale of the data problem. Current radio facilities, MeerKAT (Jonas & MeerKAT Team, 2016) and the Very Large Array (VLA; Perley et al., 2011) are already running commensal SETI

surveys (e.g. Tremblay et al., 2024) and are producing a continuous stream of candidates. However, the next generation of facilities: the Square Kilometre Array (SKA; Braun et al., 2019), and the next-generation VLA (ngVLA; Di Francesco et al., 2019), when online, will increase survey speed, sensitivity, and sky coverage by orders of magnitude relative to the instruments that currently underlie Breakthrough Listen. The SKA alone will be capable of detecting an Arecibo-class radar at the distance of any star within several hundred light-years (Siemion et al., 2015), and its commensal observing modes will produce technosignature candidate streams orders of magnitude larger than anything currently processed while the ngVLA will extend this sensitivity into the northern sky with unprecedented angular resolution.

The existing heuristic pipelines were designed for data volumes that these instruments will exceed within years, not decades. **A framework that assigns calibrated posterior probabilities to individual candidates is not merely scientifically preferable, it is operationally necessary.** Without it, the community will be unable to prioritise follow-up across the candidate streams these facilities produce, and the scientific return on millions of pounds of infrastructure investment will be limited by the inadequacy of the analysis layer sitting above it. Building that layer now, and validating it on existing Breakthrough Listen data while the new facilities complete commissioning, is precisely the right moment to act.

2 Bayesian SETI Framework

2.1 Multi-Hypothesis Problem Formulation

Each candidate signal is modelled as a multi-hypothesis inference problem over four competing explanations:

- $\mathcal{H}_{\text{instr}}$ — instrumental noise
- $\mathcal{H}_{\text{RFI}}$ — radio frequency interference
- $\mathcal{H}_{\text{astro}}$ — natural astrophysical emission (primarily M dwarf flares)
- $\mathcal{H}_{\text{signal}}$ — bona fide technosignature

Each hypothesis $\mathcal{H}_k$ is characterised by a distinct parameter set θ_k , for instance, the signal hypothesis is parametrised by $\{\dot{f}, f_0, \Delta f, \dots\}$ (drift rate, central frequency, bandwidth), while the RFI hypothesis uses $\{f_0, P, \delta, \dots\}$ (frequency, power, duty cycle). The Bayesian evidence for each hypothesis is:

$$\mathcal{Z}_k = \mathcal{P}(\mathcal{D} | \mathcal{H}_k) = \int \mathcal{P}(\mathcal{D} | \theta_k, \mathcal{H}_k) \mathcal{P}(\theta_k | \mathcal{H}_k) d\theta_k, \quad (1)$$

where $\mathcal{P}(\mathcal{D} | \theta_k, \mathcal{H}_k)$ is the likelihood and $\mathcal{P}(\theta_k | \mathcal{H}_k)$ is the prior encoding domain knowledge about plausible parameter values. The posterior probability over hypotheses follows from Bayes' theorem:

$$\mathcal{P}(\mathcal{H}_k | \mathcal{D}) = \frac{\mathcal{P}(\mathcal{D} | \mathcal{H}_k) \mathcal{P}(\mathcal{H}_k)}{\sum_j \mathcal{P}(\mathcal{D} | \mathcal{H}_j) \mathcal{P}(\mathcal{H}_j)}. \quad (2)$$

Pairwise model comparison uses the Bayes factor $\text{BF}_{ij} = \mathcal{Z}_i / \mathcal{Z}_j$, providing a calibrated, interpretable score on the Jeffreys scale. This distinguishes the present framework from existing ML pipelines: rather than a binary accept/reject decision, it returns a full posterior probability distribution over competing physical explanations for every candidate.

2.2 Likelihoods and Priors

Likelihoods are constructed from a parametric family motivated by known properties of artificial narrowband transmitters, including drifting carriers, narrowband spectral structures, and simulated RFI. Physically motivated priors are assigned to each hypothesis:

- $\mathcal{H}_{\text{signal}}$: Astrophysically motivated drift rates (Sheikh et al., 2019).
- $\mathcal{H}_{\text{RFI}}$: Earth-based transmitter characteristics including power levels, frequencies, and duty cycles, drawn from the RFI environment characterisation of Saide & Garrett (2026).
- $\mathcal{H}_{\text{astro}}$: M dwarf flare properties constraining burst duration and spectral characteristics (Wandia et al., 2025) as M dwarfs constitute the majority of stars in the solar neighbourhood (Tarter et al., 2007).

- $\mathcal{H}_{\text{instr}}$: Gaussian noise model encoding receiver specifications, gain fluctuations, and system temperature drifts (Thompson et al., 2017).

The posteriors are evaluated from the evidence integrals via nested sampling (Skilling, 2004; Buchner, 2023) using the open-source packages DYNESTY (Speagle, 2020). In Year 1 this parametric approach is used directly and in Year 2, it is superseded by the simulation-based method described in the following section.

2.3 Simulation-Based Inference

The central obstacle to principled Bayesian inference in radio SETI is the absence of a tractable analytical likelihood. Unlike gravitational-wave astronomy, where signal waveforms are calculable from general relativity, no physical theory constrains the form of technosignature emission. Any analytical likelihood model is therefore an assumption that will penalise signals outside the assumed family, a fundamental problem for a search whose whole premise is that we do not know what we are looking for.

Simulation-based inference (SBI; Cranmer et al. 2020) sidesteps this problem entirely. The key insight is that computing the likelihood analytically is not necessary: it is sufficient to be able to *simulate* data that the model would produce. A neural network trained on large numbers of simulated examples can then learn to discriminate between hypotheses directly from data. SBI encompasses several approaches depending on what the network is trained to approximate: neural likelihood estimation (NLE) learns the likelihood directly, neural posterior estimation (NPE) learns the posterior, and neural ratio estimation (NRE) learns the likelihood-to-evidence ratio, which is equivalent to the Bayes factor between hypotheses. This last flavour, NRE, is the most natural fit for the multi-hypothesis classification problem posed here.

For Breakthrough Listen, the simulator already exists. SETIGEN (Brzycki et al., 2022) generates physically motivated synthetic BL spectrograms, drifting narrowband carriers, RFI, and instrumental noise, and can inject these into real observational backgrounds. The most demanding component of an SBI pipeline for BL is therefore already in place. The neural network is trained on SETIGEN output, validated against synthetic injections with known ground-truth labels, and then applied to real BL candidates to produce a posterior probability over all four hypotheses. I note the parametric framework in the preceding section serves a dual purpose: it establishes the formal Bayesian machinery and produces a validated baseline against which the SBI framework can be benchmarked.

SBI is not an experimental technique. It has become a standard tool in astrophysics and cosmology, with robust open-source infrastructure available for immediate use. The SBI package by Tejero-Cantero et al. (2020) and the pipeline by Ho et al. (2024) both provide production-ready implementations and have been applied to problems including gravitational wave parameter estimation (e.g. Emma & Ashton, 2026), cosmological inference from large-scale structure (e.g. Jalali Kanafi & Movahed, 2026), and galaxy photometric fitting (e.g. Harvey et al., 2026). In the gravitational-wave community specifically, SBI has been used for population-level hierarchical inference across catalogues of events (Leyde et al., 2024), directly paralleling the population analysis described in Section 3. The specific application of neural Bayes factor estimation to Bayesian model comparison, the core computational task of this project, has been demonstrated on cosmological data by Jeffrey & Wandelt (2024) and Spurio Mancini et al. (2023), and the method by Srinivasan et al. (2024) provides a complementary approach to evidence estimation from posterior samples that can be used as a cross-check. The present project therefore adapts an established and well-validated methodology to a new domain, rather than developing novel algorithmic foundations.

3 Population-Level Inference: Pathway to Future Work

The posterior probabilities produced by this framework are not only useful for ranking individual candidates, they are the natural foundation for a population-level question that SETI has never been able to answer rigorously: given everything we have observed and not observed, what limits can we place on the abundance of transmitting civilisations in the Galaxy? Grimaldi & Marcy (2018) showed that under the simplified assumption of a spatially uniform distribution of isotropic emitters, the expected number of technosignature signals crossing Earth at any given time can be related to transmitter density which in principle can be inferred from survey data. The framework developed here is designed to make that observation possible in practice.

Current null results are reported as effective isotropic radiated power upper limits on individual targets. They cannot be combined across surveys in a statistically principled way, and they carry no information about how close the search came to a detection. The calibrated posteriors this framework produces can be aggregated across the full Breakthrough Listen candidate catalogue to derive a direct probabilistic constraint on galactic transmitter abundance, converting decades of null results into a meaningful bound rather than a list of non-detections. This population analysis is the natural successor to the present two year programme, not a component of it.

4 Deliverables and Timeline

The framework is positioned as a downstream complement to existing ML pipelines, not a replacement. The two-year programme produces two peer-reviewed papers and a fully open-source codebase.

Year 1 : Parametric Inference Framework

- Formulate the multi-hypothesis Bayesian framework with parametric signal models and physically motivated priors for all four hypotheses.
- Implement evidence computation via nested sampling (DYNESTY / ULTRANEST) and validate against synthetic SETIGEN injections with known ground-truth labels.
- Characterise the training requirements and distribution shift between simulated and real RFI using Breakthrough Listen archival data.
- **Paper 1** (target: *MNRAS*): “Bayesian Multi-Hypothesis Inference for Radio Technosignature Candidates: Parametric Framework and Validation.”

Year 2 : Simulation-Based Inference

- Train and validate the SBI network on SETIGEN-generated training sets across all four hypothesis classes; validate calibration via reliability diagrams and synthetic injection recovery.
- Deploy the SBI framework on BL archival survey data; re-score existing candidate catalogues including BLC-1.
- Characterise the information gain over the parametric Year 1 framework; quantify reduction in false positive rates on held-out test sets.
- Release a fully open-source Python package with documentation, trained model weights, and worked examples for community adoption.
- **Paper 2 submitted** (target: *MNRAS*): “Simulation-Based Bayesian Technosignature Classification.”

5 Conclusion

Seven decades of radio SETI have produced the most stringent upper limits in the history of the search, and no confirmed detection. Those null results are scientifically meaningful, but only if they can be expressed as probabilistic constraints rather than threshold statements. The pipeline currently underlying Breakthrough Listen was designed to find needles; it does not quantify how confident we are that a given candidate is not a needle, nor does it produce the calibrated posteriors that would make a genuine detection unambiguous.

The gravitational-wave community faced a structurally identical problem and solved it by building better inference. Bayesian analysis now underpins every GW detection pipeline. This proposal transplants that machinery into radio SETI, with simulation-based inference to handle the absence of a physical waveform model. The result is a framework that assigns every candidate a principled, calibrated probability of being a genuine technosignature, converts null results into quantitative posterior constraints, and is designed from the outset to feed into the population-level analysis that will ultimately allow the community to place the first rigorous probabilistic bound on galactic transmitter abundance.

My Background

I have training in Bayesian statistical inference, providing direct methodological grounding for the framework described in this proposal. While I am actively developing my software development skills, the computational core of the project draws on mature, well-documented open-source packages, which are designed for use by domain scientists rather than requiring bespoke development. My primary technical contribution is therefore in the statistical and physical modelling layer, where my background is strongest. I present a simple toy model demonstrating how this approach is implemented

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¹https://github.com/kelvinwandia/bayesian_techsignatures/blob/main/bayesian_demo.ipynb

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